

- 3.40 The block diagram of an autopilot designed to maintain the pitch attitude θ of an aircraft is shown in Fig. 3.65. The transfer function relating the elevator angle δ_e and the pitch attitude θ is

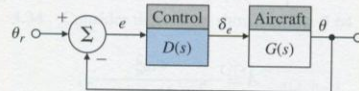
$$\frac{\theta(s)}{\delta_e(s)} = G(s) = \frac{50(s+1)(s+2)}{(s^2+5s+40)(s^2+0.03s+0.06)},$$

where θ is the pitch attitude in degrees and δ_e is the elevator angle in degrees. The autopilot controller uses the pitch attitude error e to adjust the elevator according to the transfer function

$$\frac{\delta_e(s)}{E(s)} = D(s) = \frac{K(s+3)}{s+10}.$$

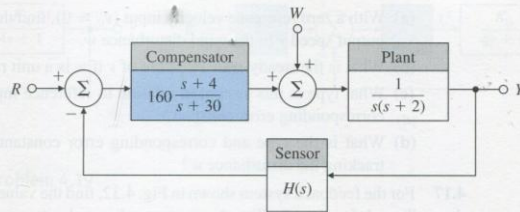
Using MATLAB, find a value of K that will provide an overshoot of less than 10% and a rise time faster than 0.5 sec for a unit-step change in θ_r . After examining the step response of the system for various values of K , comment on the difficulty associated with making rise time and overshoot measurements for complicated systems.

Figure 3.65
Block diagram of
autopilot



- 4.14 A compensated motor position control system is shown in Fig. 4.29. Assume that the sensor dynamics are $H(s) = 1$.

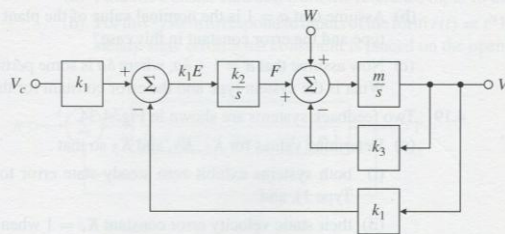
Figure 4.29
Control system for
Problem 4.14



- Can the system track a step reference input r with zero steady-state error? If yes, give the value of the velocity constant.
- Can the system reject a step disturbance w with zero steady-state error? If yes, give the value of the velocity constant.
- Compute the sensitivity of the closed-loop transfer function to changes in the plant pole at -2 .
- In some instances there are dynamics in the sensor. Repeat parts (a) to (c) for $H(s) = \frac{20}{(s+20)}$ and compare the corresponding velocity constants.

- 4.16 One possible representation of an automobile speed-control system with integral control is shown in Fig. 4.31.

Figure 4.31
System using integral
control



- With a zero reference velocity input ($v_c = 0$), find the transfer function relating the output speed v to the wind disturbance w .
- What is the steady-state response of v if w is a unit ramp function?
- What type is this system in relation to reference inputs? What is the value of the corresponding error constant?
- What is the type and corresponding error constant of this system in relation to tracking the disturbance w ?